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Network-based communication session copilot

Abstract

A system for providing a personalized assistant within a network-based communication session includes a processor and a memory storage device storing instructions. The system determines when a first communication session participant joins the network-based communication session after a threshold duration of time subsequent to the start time of the session. Upon determining the first participant has joined, the system obtains content associated with the session and creates request data for a pre-trained generative language model. The request data includes an instruction requesting a predetermined number of suggested utterances not present in the content, each utterance relating to one or more topics corresponding to the content. The system transforms the request data to a command based on a command template and provides the command to the generative language model. The system receives a response from the model, including the predetermined number of suggested utterances, and presents them to the communication session participant in a graphical user interface while the session is in session.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/448,624, filed on Feb. 27, 2023, and titled “NETWORK-BASED COMMUNICATION SESSION COPILOT,” the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) Embodiments relate to the use of generative language models, (e.g., large language models, or “LLMs”) to improve network-based communications such as network-based communication sessions. Further embodiments pertain to using machine-learned language models to provide a

personal assistant to communication session participants.

BACKGROUND

(2) Network-based communication sessions, such as network-based meetings, allow users to interact with people in remote locations. In addition to providing voice and video capabilities, network-based communication sessions also allow users to exchanged text-based messages, share content, applications, screens, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

(2) FIGS. 1-11 illustrate various graphical user interfaces (GUIs) of a network-based communication application engaged in a network-based communication session according to some examples of the present disclosure.

(3) FIG. 12 illustrates a network-based communication session environment according to some examples of the present disclosure.

(4) FIG. 13 illustrates a logical diagram of a copilot system according to some examples of the present disclosure.

(5) FIG. 14 illustrates a GUI of a collaborative copilot system according to some examples of the present disclosure.

(6) FIG. 15 illustrates a flowchart of a method for providing a personalized assistant (e.g., a copilot) within a network-based communication session.

(7) FIG. 16 illustrates a flowchart of a method for providing a personalized assistant (e.g., a copilot) within a network-based communication session, by creating a summary of summaries, according to some examples of the present disclosure.

(8) FIG. 17 illustrates a flowchart of a method for providing a personalized assistant (e.g., a copilot) within a network-based communication session by creating a rolling summary to some examples of the present disclosure.

(9) FIG. 18 illustrates a flowchart of a method of querying a transcript of a network-based communication session according to some examples of the present disclosure.

(10) FIG. 19 illustrates a flowchart of a method for caching a predicted query response according to some examples of the present disclosure.

(11) FIG. 20 shows an example machine learning module according to some examples of the present disclosure.

(12) FIG. 21 is a block diagram illustrating an example of a machine upon which one or more embodiments may be implemented.

DETAILED DESCRIPTION

(13) Network-based communication sessions, such as network-based meetings, especially those involving many participants are often rapid-fire environments where participants may struggle to keep up with the action. One errant thought, one distraction, one screaming child in the background may distract a participant long enough to miss important parts of the network-based communication session. While a participant may ask for clarification, this is often disruptive or embarrassing, especially for participants who may be shy.

(14) Network-based communication sessions may also be very long, often several hours, to the point that it may be difficult for participants to remember decisions made, opinions of participants, pros and cons of ideas presented, and the like. Participants may thus struggle to accurately

remember or understand decisions that have been made, action items that are outstanding, open questions, and other aspects of the network-based communication session. In addition, participants may have differing recollections of what was said, or what occurred, during the communication session, such that accurate reconstruction may be impossible.

(15) In addition, participants may experience moments of the communication session where the participant's progress appears “stuck.” That is, the participants may not have a clear path forward toward achieving their goals for the communication session. This may be as a result of disagreements between participants, uncertainty about decisions to-be made, and the like. In addition, late-arriving participants may struggle to catch up with what has already been decided or discussed.

(16) Finally, after the meeting, participants may have difficulty remembering what happened during the communication session. Additionally, individuals with scheduling conflicts that were unable to participate may have difficulty understanding decisions that were made, topics that were discussed, and the like.

(17) One tool that can assist users during network-based communication sessions with these problems is the live transcript. The network-based service translates voice conversation data in real time to generate a voice transcript of the communication session. Despite this transcript, it may take quite some time for the user to scroll back to find the information they are searching for. In a real-time and dynamic communication session, the user may have to take their focus off the communication session to find the information they are looking for which may result in them missing even more content. In addition, the live transcript traditionally does not include content shared during the communication session, thus the transcript may be either incomplete or lack important context.

(18) Disclosed in some examples are methods, systems, and machine-readable mediums for providing a network-based communication session copilot. The network-based communication session copilot may serve as a personalized assistant to a communication-session participant that provides information and advice about the network-based communication session. For example, the copilot may answer participant questions during or after the network-based communication session about the session such as about shared content such as the transcript, chats, files, screen sharing; previous communications such as emails, chats, documents, and content from previous communication sessions; and the like. Example tasks include summarization of the communication session or portions thereof, identification and summarization of different topics in the communication session, list of participant opinions, open questions, concrete questions on content shared or discussed during the communication session, specific questions about participants, and the like. In some examples, the copilot may provide information about the communication session after the communication session.

(19) In some examples, participants may interact with the copilot in a number of ways. For example, a participant may ask the communication session copilot free-text, natural language questions and receive natural language responses. In other examples, the copilot may anticipate the questions of participants. For example, the copilot may recommend, from a prespecified list of questions, one or more of the most relevant questions to ask based upon the current communication session content, the role of the user in the session, previous questions and answers of the participant, and/or the like. The communication session copilot may suggest follow-up questions based on the communication session content, the question, and previous answers. In some examples, the copilot may proactively initiate an answer to a question the user has not yet asked. In some examples, the copilot may scan the meeting transcript periodically and prompt a participant. For example, by stating “John is asking you about” a particular topic.

(20) In some examples, the communication session copilot may personalize the answers based upon the user's style. For example, the copilot may determine a user's style and/or interests from the phrasing used from submitted free text queries (in the current communication session or

previous sessions of the user), what the user says during the communication session (e.g., from the communication session transcript), and other contextual signals. Example style changes include providing more concrete answers if the user prefers more concrete results and if the user has more doubts, the answer could provide different options. For example, the copilot may learn what is relevant and interesting to the user across different meetings and apply those lessons to providing a relevant answer. Other example styles may include short and concise answers vs. detailed answers; answer formatting (table vs bullets vs paragraphs); quoting the transcript vs having a summary, and/or formal language or casual language.

(21) Example information provided may include a summary of what has been discussed so far, decided-upon action items, information on participants (e.g., such as a current speaker), determine unresolved questions, determine varying opinions, list main ideas discussed, and the like. The communication session copilot may automatically summarize a communication session for a late-joining participant (e.g., with or without a participant requesting it), provide suggestions for driving the communication session forward, help users break the ice (e.g., provide stories, jokes, or the like), highlight different perspectives, suggest polls when asking questions with choices, and the like. Example suggestions for driving the communication session forward include providing questions that participants can ask-which may be leading questions, provide pros and cons of a particular decision point, enrich the discussion with world knowledge and different perspectives, and the like. In addition, the copilot may identify when the conversation strays from a submitted agenda and provide prompts for users to get the communication session back on target.

(22) By way of example, when a meeting participant is later in joining an in-session meeting, the copilot may provide a catch-up summary of the portion of the meeting that was missed. For instance, the system may determine a meeting participant is late if the meeting participant joins the meeting at some time after the start of the meeting, where the difference between the start of the meeting and the time at which the participant joined is greater than some predetermined threshold (e.g., five minutes, ten minutes, 20 minutes, and so forth). Upon detecting that a meeting participant was late to join, the system may automatically generate a prompt or text input for a generative language model, where the instruction included in the prompt asks for a summary of the conversation or content presented from the time the meeting began, until the time at which the participant joined. Accordingly, a live meeting transcript representing the window of time may be provided as context, in the prompt, or via consecutive sequential prompts, to the generative language model. The resulting output of the generative language model is then presented to the meeting participant in a user interface, allowing the meeting participant to quickly “catch-up” on what has occurred in the meeting, prior to the participant joining.

(23) In some examples, the copilot may be specific and private to each participant. That is, each participant may have a private instance of copilot that may be isolated from other participants. In these examples, questions a participant asks and answers provided may not be visible to other participants. In other examples, a collaborative copilot may be provided instead of, or in addition to the private copilot. The collaborative copilot may be a shared experience for all participants. Example collaborations include notification of users when a topic is discussed, or when a different opinion is discussed; identify when the conversation goes off the agenda; highlight different perspectives; and suggests polls when asking questions with choices; question and answer that all can see; allowing users of the communication session to edit the answers; capturing action items and notes; and the like. When users edit answers provided by the communication session copilot, the system may adapt the model so that future answers learn from the explicit feedback given by the users. In some examples, both a collaborative copilot and a private copilot may be provided.

(24) In some examples, the copilot may utilize the communication session transcript, chats, files, and/or any audio or video shared during the current communication session. In some examples, the copilot may utilize communication session transcript, chats, files, and/or any audio or video of previous communication sessions that are related to the current communication session (e.g.,

previous recurring communication sessions). In still other examples, the copilot may utilize participant emails, files, and other content. The copilot may utilize one or more machine-learned language model to provide the above disclosed functionality. Example machine-learned language models include generative language models, frequently referred to as large language models (LLMs), such as a Generative Pre-Trained Transformer (GPT) model.

(25) A generative language model is a type of model that is trained to generate coherent and contextually relevant text. A generative language model “learns” the statistical patterns and dependencies of language by analyzing vast amounts of training data. The model can then generate new text based on the patterns the model has observed. The training process for a generative language model typically involves the following steps: Data Collection: Gather a large corpus of text data from diverse sources such as books, articles, websites, or other textual sources. The data should cover a wide range of topics and exhibit the language patterns and structures that the model is to learn. Text Preprocessing. Clean and preprocess the text data by removing irrelevant content, special characters, and formatting inconsistencies. Tokenization is commonly performed, breaking the text into individual words or subword units. Training Architecture: Select a suitable architecture for training the language model, such as a recurrent neural network (RNN) or transformer-based models like GPT (Generative Pre-trained Transformer). These architectures are designed to capture contextual dependencies and generate coherent text. Training Objective: Define the training objective, which is typically to maximize the likelihood of predicting the next word or sequence of words in the training data. This is achieved through a process called maximum likelihood estimation (MLE). Model Training: Train the language model on the preprocessed text data by presenting it with input sequences of words and optimizing its parameters to minimize the difference between predicted and actual next words. The optimization is done using gradient-based techniques such as backpropagation and stochastic gradient descent. Fine Tuning: Once the model has been trained, the pre-trained generative language model can be further fine-tuned on specific downstream tasks or domain-specific datasets to enhance its performance in those specific areas.

(26) In some examples, a model may be fine-tuned using a supervised learning technique that involves using as training data annotated or labeled communication session or meeting transcripts. For example, a communication session transcript may include chronologically ordered content items with each content item representing a communication (e.g., a spoken statement) of a meeting participant during a meeting. The meeting transcript may also include other communications, such as those that occurred via a chat function or feature, and in some instances, content that has been shared during the meeting via a collaboration feature or function. Accordingly, each content item may also include information identifying the person who made the communication, and the time during the meeting at which the communication was made. For use as training data, some content items, or portions of content items, may be annotated or labeled to identify those content items as representing specific concepts, such as questions, instructions, to-do or follow-up task-based items, opinions, and so forth. Using the training data, the model can be fine-tuned to generate responses to specific instructions (e.g., as included in a text input or prompt). For example, a prompt or text input to the model may include an instruction or request that the model generate a list of questions. Specifically, the instruction may request that the model generate some number of questions that have not yet been asked by a meeting participant. Alternatively, the instruction or request may ask that the model generate some number of questions that have been asked, but not answered. The specific text representing the instruction or request will of course vary, and may itself be iteratively optimized via several rounds of testing and tweaking.

(27) In some examples, multiple models may be utilized. For example, an intermediate model may process user input (e.g., the user query), the communication session transcript, the role of the user in the communication session, communication session metadata (time of the communication session, title of the communication session, list of participants, communication session location, communication session agenda, and the like), transcripts of past relevant communication sessions,

and the conversation history of the participant with the copilot. Based upon these inputs, the intermediate model may then generate one or more prompts to the LLM. The answers from the LLM may be processed and then provided back to the user.

(28) In some examples, the intermediate model may add to the prompts to the LLM a question to determine possible follow up questions or queries. In parallel to providing the response to the original query to the user, the intermediate model may query the LLM on the answers to the follow up questions. In this way, the system predicts user questions and pre-caches the answers to avoid additional latency.

(29) In some examples, input to large language models such as GPTx are limited in size. Such input is typically in the form of a textual context and an instruction. Accordingly, handling long context is nontrivial. Transcripts of short network-based communication sessions may fit into the input constraints of these models. However, long network-based communication sessions, such as those exceeding sixty minutes have transcripts that far exceed the input limit to these models. This makes it non-trivial to generate different types of summaries, and, more generally and more dynamically, executing user queries in free text to extract information and answer questions about the communication session.

(30) In some examples, the copilot may submit additional meeting content such as videos, shared documents and the like to the LLM. In some examples, the intermediate model may determine relevant information and then provide those results to the LLM together with the transcription. In other examples, the copilot may use the intermediate layer to identify the relevant sections from the files/chat etc. and feed the original sections to the model as text together with the transcription and query to the model. In still other examples, the copilot may add the document to the model prompt if it fits into the input or make several prompts to the model if it doesn't fit and combine the results with a final prompt (e.g., as detailed below for the transcripts). In some examples, to determine relevant information from additional context, the intermediate model may utilize deep learning AI techniques such as convolutional neural networks (CNN).

(31) In some examples, the communication session copilot may solve these problems by utilizing an iterative submission process to the LLM. In some examples by using a summary of summaries where context info (e.g., the transcript and/or shared content) may be partitioned into sections. Each section is then summarized to create sub-summaries. The sub-summaries are then summarized to create an overall summary. In other examples, a rolling summary may be used. Starting from an initial summary, the summary is iteratively extended to cover each successive section until the entire context is covered. For user queries on the text, the query may be used to create a rolling summary that includes all the relevant details for the query and the query may then be applied on the completed summary.

(32) SUMMARY OF SUMMARIES

(33) To process large transcripts, a first embodiment partitions the transcript T of the network-based communication session into N sections {T.sub.1, T.sub.2, . . . , T.sub.N}. A summary S is created of each section to create N summaries {S.sub.1, S.sub.2, . . . , S.sub.N}. An overall summary S.sub.Output is created that summarizes all of the individual summaries. In some examples, the communication session copilot may adjust the style of the summaries (e.g., make them short) and their content and form (e.g., include only action items or create a summary in table form) by including elaborate instructions in the prompts to the LLM. One of the major advantages of this summary of summaries method, is that the first phase of creating section summaries, can be easily parallelized, such as by using a map-reduce framework. Nonetheless, section summaries may be generic and high-level, or wrong, apparently due to lack of context. In some examples, to solve the problem of lack of context, the system may overcome this by creating an overlap in the sections. Thus, for example, a portion of T.sub.1 overlaps T.sub.2. In some examples, an ending portion of T.sub.1 overlaps a beginning portion of T.sub.2. In some examples, the summary of summaries may be parallelized. That is, each of the N summaries may be created simultaneously or near-

simultaneously and then combined in the final result. In some examples, the overlap may be a particular number of sentences, such as 6-8.

ROLLING SUMMARY

(34) To overcome some of the challenges with the summary of summaries approach, in some examples, a rolling summary may be used. As with the summary of summaries, the transcript may be divided into N sections $\{T_{\text{sub.1}}, T_{\text{sub.2}}, \dots, T_{\text{sub.N}}\}$. A summary $S_{\text{sub.1}}$ is created from the first section $T_{\text{sub.1}}$. Then, given $S_{\text{sub.1}}$ and given the transcript of the second section $T_{\text{sub.2}}$, the summary is extended to also cover section 2. The process is repeated iteratively extend the summary to cover $T_{\text{sub.3}}$ and so on until $T_{\text{sub.N}}$ is processed and the summary is completed. This way, via a rolling summary, each section generally gets the entire backward context in compressed form. Note that the partitioning of the transcript can be done on-the-fly—in each step, considering the actual size of the rolling summary. Also, and as previously noted, the prompts may be modified and extended in several ways; e.g., the style, content, and form of the summary can be easily adapted.

Queries in Free Text

(35) The summaries described above may be useful to get some details on the communication session, but they may not always be helpful in answering many of the more specific questions about the communication session. In order to answer user queries in free text, the system may utilize another technique. For short transcripts, the prompt to the LLM may include the transcription and/or other context; and the user query with an instruction to answer the user query based upon the transcript. In some examples, the instructions may be augmented with additional system instructions.

(36) In case the context (e.g., the transcript) is too long, the system may not be able to use the LLM to directly query it. Nonetheless, the system can use the rolling summary approach in order to create an ad-hoc summary of the text that is focused on the provided query, and then use that ad-hoc summary to answer the query. For example, the system may first prompt the language model to summarize a first part of the communication session and including all the details from the communication session that are needed or may directly help to later provide an answer to the query. The LLM may be directed to refrain from mentioning the query itself, but to still include the relevant details. After the summary is produced using the previous query, the copilot may issue a second command that includes the user's query, the summary of a first part, and a transcript of a next part. The LLM is instructed to create an extended summary covering the first part of the communication session and the next part of the communication session, including all the details (if any) that are needed or may directly help or later provide an answer to the user query. This prompt may be used iteratively until the entire transcript is consumed. Finally, to execute the query given a query-aware summary of the entire communication session, a prompt to the LLM may be submitted that includes the query-aware summary, and the user query. The prompt asks the LLM to answer the user query based upon the summary of the communication session.

(37) In other examples, to answer a query, each transcript segment is not summarized but rather, the copilot asks the LLM to answer the original question (query) on each segment independently, and then the responses are combined. In these examples, this may utilize a map-reduce framework to parallelize the process. In some examples, when using this method, the query Q may be converted into another query Q_2 that is used in and contains more information than Q , and better supports turning the several responses into one. For example, turning the query Q ="Is alternative A better than B?" (which is a yes/no question that is hard to "reduce") into "What are the pros and cons of A and B" (which is much easier to reduce).

(38) In still other examples, the summary of summaries approach may be modified. For example, a query-aware summary of each portion of the context is constructed (the summary of each part is instructed to include all the necessary details to answer the query). These individual summaries are then summarized into one final summary and this final summary is used to answer the query by

prompting the model.

(39) In some examples, the copilot may compress the context by creating a concise summary that removes redundancies and repetitions but leaves relevant details. For example, by removing repetitions, mumblings, fill words (“ah,” “um”, etc. . . .). This may be done by utilizing the LLM to remove these irrelevant details. In other examples, a list of irrelevant phrases may be used to remove them from the context.

(40) Alternatively, the system may generate a few compressed texts, each covering a different aspect (e.g., topic-such as technology, management, etc. . . .), and the model may be used to select the relevant one given the query. To control the experience, and the “character” of the model, more context may be provided in the prompts.

(41) By leveraging these techniques the copilot may provide assistance to participants of a communication session. For example, if a participant joined late to a session, they may be reminded to catch up on what they might've missed using the copilot. The copilot may show a topic-based summary allowing them to catch up quickly without needing to interrupt the ongoing conversation.

(42) As another example, when a communication session is about to end, participants may be automatically asked to wrap up the communication session and notes will be automatically generated on their behalf to highlight key topics and action items. As yet another example, if a conversation is going off agenda, participants may get a prompt that conversations should be pulled back to an item on the agenda. In these examples, the agenda may be submitted by an organizer of the communication session. In some examples, the copilot may proactively detect divergent opinions, emotions/tensions, and/or whether goal is achieved in a communication session and notify participants with suggested actions to improve session effectiveness. The copilot system may use prompts (free text, pre-defined and suggested by AI) to enable participants to reach shared understanding on what's discussed.

(43) This real-time analysis of conversations in a communication session may use one or more of transcribed speech, chat messages, agendas, attached documents, title, other communication session artifacts, with crafted prompts that produces relevant and accurate results, on top of a large language model.

(44) For participants that had to leave the communication session early, they can ask the communication session copilot to catch up on what they missed after they left and get more context on action items assigned to them. Likewise, for participants who didn't attend the communication session, they can see notes generated by AI and ask the communication session copilot to get deeper context on what's discussed, without needing to watch the recording or read the transcript. If a user was on vacation for x days, they can ask the communication session copilot to recap communication sessions they missed during that time, and highlight what's important for them as well as action items, without needing to go back to notes/recording/chat to catch up.

(45) The disclosed methods, systems, and machine-readable-mediums thus solve the technical problems of managing information in network-based communication sessions by improvements in interactions and usability and user efficiency. In addition, the disclosed techniques solve problems related to limited input sizes of LLMs through technical iterative processing solutions. In addition, the proposed solutions solve technical challenges of utilizing an LLM on live, dynamic content, such as a network-based communication session through technical solutions of iterative processing, intermediate models, and/or the like. In some examples, the proposed techniques utilize specific rules or models to generate prompts to the LLM that increase the reliability and usability of the information from the LLM and remove the human judgment from the prompts to the LLM to create more consistent results.

(46) FIG. 1 illustrates a GUI **100** of a network-based communication application providing in a network-based communication session according to some examples of the present disclosure. In the example of FIG. 1, the network-based communication session is a network-based meeting. The GUI includes a toolbar **105** with options for leaving the communication session; sharing content;

muting or unmuting a microphone; enabling or disabling video; starting other applications; changing a view; leaving a reaction; raising a virtual hand; viewing a list of people; viewing a chat; viewing the copilot; and the like. In the example of FIG. 1, the copilot view is selected. This brings up the copilot pane **110**. The copilot pane **110** includes a suggested query with a selectable control **115** to execute a query to “catch me up on what's been talked about so far.” In addition, a text box **120** allows users to enter custom natural language queries, and a selectable control **125** that displays additional suggested queries. Custom, free text prompts to the large language model may allow participants to ask the copilot any question about the meeting in progress, past meetings, or the like.

(47) FIG. 2 illustrates a GUI **200** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. FIG. 2 illustrates the GUI **100** after the participant has selected the selectable control **115**. The copilot pane **210** shows a natural language answer **205**, as well as a plurality of selectable controls **215** that allow the participant to submit one or more suggested queries.

(48) FIG. 3 illustrates a GUI **300** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. FIG. 3 illustrates the GUI **100** after the participant has selected the selectable control **325** to show the suggested prompts. The GUI **300** displays a list of selectable controls **330** with various suggested queries.

(49) FIG. 4 illustrates a GUI **400** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. FIG. 4 illustrates a GUI **400** with a copilot pane **410**. Copilot pane **410** includes a variety of queries and answers. For example, a first answer **445** (the query of which is not shown (e.g., in some examples, it may be visible by scrolling upward), with a second question **440** and second answer **442**. The second answer **442** may be scrollable such that the entire answer may be accessed by scrolling up and down within the response box. Selectable controls **420** show various suggested prompts. Suggested prompts selectable control **425** may show additional selectable controls for additional suggested prompts.

(50) FIG. 5 illustrates a GUI **500** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. The GUI **500** shows a prompt of “who is speaking now” at **540**. The response **542** includes a name as well as information about the current speaker. Furthermore, context information about the speaker related to the current communication session may be shown. For example, in FIG. 5, not only is the user's name and company present, but the fact that they are a “guest” speaker and also what the speaker is talking about. This information may be obtained by participants introducing the speaker, by directory services, or the like.

(51) FIG. 6 illustrates a GUI **600** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. GUI **600** is a continuation of GUI **600** where the user has asked “what questions can I ask Nicholas?” at **640**. The copilot has responded at box **642** with possible questions.

(52) FIG. 7 illustrates a GUI **700** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. GUI **700** is a continuation of GUI **700** where the user has asked “What are the decisions made?” at **740**. The copilot has responded at box **742** with decisions that the team has made.

(53) FIG. 8 illustrates a GUI **800** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. GUI **800** is a continuation of GUI **800** where the user has asked the copilot to “capture action items” at **840**. The copilot has responded at box **842** with action items.

(54) FIG. 9 illustrates a GUI **900** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. GUI

900 shows a table of pros and cons **940** for each idea expressed during the communication session. The pros and cons are taken from the content of the network-based communication session such as the pros and cons discussed by participants.

(55) FIG. **10** illustrates a GUI **1000** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. GUI **1000** shows a table of pros and cons **1040** for each idea expressed during the communication session. The user has selected a selectable control that brings up a menu **1045** that allows the participant to copy the table to a notes section of the network-based communication application, send the table via email, open it in a word processing application, copy it to a clipboard, or the like.

(56) FIG. **11** illustrates a GUI **1100** of the network-based communication application engaged in a network-based communication session according to some examples of the present disclosure. The GUI **1100** is of a notes section of the network-based communication application that presents notes of the network-based communication session. The notes section **1120** may include manually taken notes, notes produced by the copilot, and the like. Shown in the notes section **1120** is the pros and cons table from FIG. **10**. Additionally, communication session goals, key topics, and varying opinions may be populated by the copilot. In some examples, the notes section **1120** may utilize a template, where the fields of the template are auto-filled by copilot as the communication session progresses. The notes section **1120** may also be edited by participants collaboratively.

(57) FIG. **12** illustrates a network-based communication session environment **1200** according to some examples of the present disclosure. Participant computing device A **1212** and participant computing device B **1210** may communicate with a network-based communication service **1214**—e.g., over a computing network. The network-based communication service **1214** may provide the network-based communication session by receiving audio, video, content, screen sharing, and other data from each participant computing device and forwarding that data to other computing devices of other participants.

(58) In some examples, copilot commands and/or queries may be processed locally on the participant computing devices. That is, the models such as the LLMs may be downloaded to the participant computing devices locally. In other examples, the models may be within the network-based communication service. Commands, queries, prompts, and other requests of the copilot may be sent from the participant computing devices to the network-based communication service **1214**. In some examples, the models are at the network-based communication service **1214**. In other examples, one or more models are located at a different network-based service such as a network-based language model service **1216** that is reachable over a network from network-based communication service. The commands, queries, prompts, and other requests of the copilot may be forwarded to the network-based language model service **1216**. In some examples, the network-based communication service **1214** may host an intermediate model, such as that described in FIG. **13**.

(59) FIG. **13** illustrates a logical diagram of a copilot system **1300** according to some examples of the present disclosure. The copilot component **1328** may be part of the network-based communication service (such as network-based communication service **1214**), or the network-based language model service (such as network-based language model service **1216**) in some examples and intermediates between the communication session participant and the LLM **1330**. For example, a user's query **1320** (either a free text query, or one or more prespecified queries) may be submitted to the copilot component **1328**. Additionally, a current live transcript **1310** of the communication session, user data **1312**, other communication session metadata **1314** (participant locations, number of participants, time of the session, current duration of the session, communication session agenda, participant information, session title, and the like), transcripts of past relevant sessions **1316** (e.g., if the communication session is a recurring communication session other past communication sessions of the series; other communication sessions with similar or a same subject; other communication sessions with similar or a same title and/or agenda, or the

like), a conversation history between the user and the copilot in this session and/or other relevant sessions **1318**, media shared during the communication session **1319** (including files, screen sharing, videos, communication session chat history between participants, and the like), and the like. In some examples, user data **1312** may include a role of the user in the communication session (e.g., organizer, presenter, leader, manager, or the like), a name of the user, a title of the user, an organization of the user, and the like.

(60) These inputs may be used by the copilot component **1328** along with the user query **1320** to produce one or more model prompts **1332**. The model prompts may be generated by the copilot component **1328**. For example, the copilot component **1328** may use an intermediate model, such as intermediate model **1350** to generate the prompts. In some examples, the intermediate model **1350** may be a rule-based model that may utilize one or more rules that select one or more template model prompts that are then completed using the inputs to the copilot component **1328** and output as the model prompts **1332** to the LLM **1330**. In other examples, other than rules, other models may be used, such as for example random forests, decision forests, another LLM, or the like.

(61) In some examples, the intermediate model may utilize pre-defined templates for creating the prompt. Given the input and additional parameters as the user's question and copilot interaction history copilot may utilize a classifier which selects the best prompt for the LLM. In other examples, a generative model may use certain building blocks for the prompt based upon the inputs discussed above. This model is trained to generate a prompt given the user's input, the response of the model and a label whether this response is good.

(62) LLM **1330** may be a large language model. Examples include GPT models. The LLM **1330** operates on natural language input prompts, such as model prompts **1332** and generates a natural language response, such as response **1334**. The response may be interpreted by the copilot component **1328** to determine if additional prompts are to be issued. Additional prompts **1336** may be issued and responses **1338** may be received. Multiple rounds of prompts and responses may be issued by the copilot component **1328** to the LLM **1330** for a single user query **1320** depending on the query. For example, to generate the pros and cons for each idea, a first prompt may obtain the list of ideas, and the LLM **1330** may be queried for each idea to find the pros and cons of that idea. In addition, multiple prompts may be, for example, to produce a summary (e.g., using the summary of summaries or rolling summary techniques) and then further prompts may be utilized to apply that summary to a particular query. Once the copilot component **1328** has the requisite information to answer the query, the copilot component **1328** may send a response **1322** to the participant.

(63) In some examples, the model prompts may include prompts for the LLM **1330** to also include, in the response, likely follow up questions that the participant may have. As the copilot component **1328** delivers the response **1322**, it may prefetch answers to those suggested queries **1340** and cache the responses **1342** using a response cache **1352**. If the user asks one of those follow up questions, the answer may be served from the response cache **1352** of the copilot component **1328**. This reduces latency for these queries by pre-storing the responses and thus improves the functioning of the computing system. In some examples, the pre-stored responses may have expiration times, after which, the cached answer is cleared. This prevents stale answers that are superseded by more recent events in the network-based communication session. The response cache **1352** may be volatile or non-volatile memory.

(64) In some examples, the copilot component **1328** may produce one or more suggested queries **1324**. These suggested queries may be based upon a role of the user, the content of the communication session, and the like. The suggested queries may be created by the LLM **1330**, or the copilot component **1328**. In some examples, the suggested queries may be chosen from a prespecified list of queries. In other examples, the suggestions are based upon the communication session content, previous answers, and recent queries of the users and may be suggested queries that are not prespecified.

(65) Summary component **1354** may partition one or more portions of the inputs (such as transcript

1310, media shared during the communication session **1319**, and the like). The summary component **1354** may then produce a summary, such as by using a summary of summaries, rolling summary, or summary for a query as previously described. In some examples, the sizes of the partitions of the portions of the inputs may be created using fixed sizes and an input limit for the LLM. For example, the difference between the input limit and the prompt language may be a maximum partition size. In other examples, the partition size may be dynamically created. For example, based upon the transcript length, the copilot history length, the LLM input limitation, user query length, expected output length, metadata length (username, meeting title, meeting invitation body, etc.).

(66) In some examples, the partitions may overlap. This may assist in preserving context for techniques like the summary of summaries. The amount of overlap may be prespecified. In other examples, the amount of the overlap may be dynamically calculated. For example by dividing the meeting into semantically separated segments, then either using non-overlapping sections or using a segment which is relevant to both the previous and next segments as the overlapping section.

(67) Participants may provide user feedback **1326**, which may be edits to the responses and/or suggested questions, and the like. This feedback may be used by the copilot component **1328** to update the intermediate model **1350** and/or refine the LLM **1330**. When a user edits a response the system stores the changes and the uses the edits as additional input to the LLM. The edits are given in a different section in the prompt and then we ask the model to take these changes into account when answering the next question. This way copilot can fix erroneous utterances/words in the transcription, fix errors or emphasize specific details.

(68) FIG. **14** illustrates a GUI **1400** of a collaborative copilot according to some examples of the present disclosure. The GUI **1400** may appear on a main stage portion of the network-based communication application. In some examples, the personal copilot may also be displayed simultaneously with the collaborative copilot. For example, the collaborative copilot may take the place of user's avatars or video squares, such as shown in FIG. **8**. The collaborative copilot is a shared experience with all participants able to see and interact with a common copilot. The collaborative copilot may have a summary pane **1410** that shows a summary of goals, key topics, varying opinions, and other information about the communication session. The collaborative copilot may also include selectable graphical elements in another pane **1412** that allow participants to ask copilot questions such as “what's missing from the discussion?” “suggest new ideas related to topics discussed,” “what topics need more clarification,” “what are unresolved questions,” and other options to explore. In some examples, the functionality of the collaborative view may be a restricted functionality of the private copilot. For example, queries about people in the meeting may not be allowed in the collaborative view.

(69) FIG. **15** illustrates a flowchart of a method **1500** for providing a personalized assistant (e.g., a copilot) within a network-based communication session. At operation **1510**, the copilot may receive a query from a participant of the network-based communication session. The query may be a free-text query, a suggested query, or the like. For example, if the participant is late to the communication session, the system may suggest that the participant “catch up” with what happened by getting a summary of the communication session to that point in time. In some examples, the query may be typed by the user into an entry box, but in other examples, the query may be entered by clicking or otherwise selecting a selectable element of a GUI such as a selectable suggested query. In some examples, the query may be received from a client communication application that sends it over a network to a communication service.

(70) At operation **1515**, the copilot may construct a prompt to a language model such as a machine-learned large language model. In some examples, the prompt includes a portion of content of the network-based communication session and instructions based upon the query. The portion of content may include one or more of: a summary of a transcript (e.g., previously created by the LLM), a file shared during the network-based communication session, a communication session

agenda, a communication session title, information about participants of the network-based communication session, a conversation history between the personalized assistant and the participant, video shared during the network-based communication session, an email of the participant, or chat content. In some examples, the prompt may be constructed based upon one or more prompt templates that may be selected based upon the query and filled in with relevant communication session details. In some examples, the prompts may be constructed by a second model, such as a second machine-learned model. Example models may include a second large language model that may be of a same or different type than the large language model sent the prompt. In still other examples, the large language model may be asked to generate the prompt. (71) At operation **1520**, the copilot may submit the prompt to the machine-learned large language model. In some examples, this may include calling the machine-learned large language model locally or may include sending the prompt using an API of the machine-learned large language model.

(72) At operation **1525**, the copilot may receive a response from the machine-learned large language model. As with operation **1520** this may be a locally received response or a response received over a network.

(73) At operation **1530**, the copilot may provide a result to the participant as part of a Graphical User Interface of the network-based communication session, the result based upon the response. In some examples, the result may be the response, but in other examples, additional prompts may be submitted to the machine-learned large language model and the additional responses may be used instead of, or additionally to the response to provide the result. For example, the copilot may have logic to determine a quality of the response. If the quality is below a threshold, a prompt with new language may be utilized and the response of that prompt may be utilized. In some examples, the query of the user may require additional prompts. That is, the query may require multiple prompts and responses and each prompt and response answers a portion of the user's query. For example, to create the pros and cons list, the copilot may first query the LLM for the ideas expressed. Then for each idea, the copilot may query the pros and cons discussed for each idea. In other examples, multiple prompts may be utilized to, for example, summarize the transcript of the communication session to allow for searching the query.

(74) FIG. **16** illustrates a flowchart of a method **1600** for providing a personalized assistant (e.g., a copilot) within a network-based communication session by creating a summary of summaries according to some examples of the present disclosure. Method **1600** may be utilized to create a summary of summaries submitted, e.g., as part of the prompt in operation **1515**. In other examples, operations **1515-1525** may be a first portion of creating a summary of summaries wherein the prompt to the model requests the model to summarize a first portion of the live transcript of the network-based communication session. In these examples, the prompt at operation **1515** may request that the LLM summarize a first portion of the transcript and the response at operation **1525** is a first summary. The operations of FIG. **16** may be performed between operations **1525** and **1530** in these examples.

(75) At operation **1610**, the copilot may construct a second prompt to the machine-learned large language model, the second prompt including a second portion of the live transcript of the network-based communication session and second instructions to summarize the second portion of the live transcript. In some examples, the first and second portions of the transcript may be overlapping. Using overlapping transcripts may reduce incidents where the final summary produced by the summary of summaries approach fails to provide appropriate relevant context.

(76) At operation **1615**, the copilot may submit the second prompt to the machine-learned large language model. At operation **1620**, the copilot may receive a second response from the machine-learned large language model, the second response being for the second prompt. As noted for FIG. **15**, the prompts and responses may be sent and received locally or via a network.

(77) At operation **1625**, the system may construct a third prompt to the machine-learned large

language model, the third prompt including the response and second response and third instructions to summarize the response and second response. At operation **1630**, the copilot submits the third prompt to the machine-learned large language model. At operation **1635**, the copilot receives a third response from the machine-learned large language model, the third response being a summary of the entire communication session up to the point the request for the summary was received. At operation **1640**, the copilot determines the result based upon the third response. In some examples, the third response may be the result. In other examples, the system may utilize the third result in another prompt to the LLM—for example, issuing a query on the summary and the response to that prompt may be the result.

(78) While FIG. **16** illustrated breaking the transcript into two portions, summarizing those portions, and then summarizing the summaries, in other examples, the copilot may break the transcript into more than two portions, summarize those portions, and then summarizing those summaries. In the operations of FIG. **16**, two summary stages are utilized. A first to summarize a portion of the transcript and a second to summarize the summaries. In other examples, for very long communication sessions, additional stages may be used. For example, a first stage may summarize a large number of transcript portions. These summaries may be grouped together and then those groups summarized. The summaries of the groups may then be summarized and so on. Thus, more than two stages of summarization may be employed.

(79) FIG. **17** illustrates a flowchart of a method **1700** for providing a personalized assistant (e.g., a copilot) within a network-based communication session by creating a rolling summary according to some examples of the present disclosure. Method **1700** may be utilized to create a rolling summary submitted, e.g., as part of the prompt in operation **1515**. In other examples, operations **1515-1525** may be a first portion of creating a rolling summary wherein the prompt to the model requests the model to summarize a first portion of the live transcript of the network-based communication session. In these examples, the prompt at operation **1515** may request that the LLM summarize a first portion of the transcript and the response at operation **1525** is a first summary. The operations of FIG. **17** may be performed between operations **1525** and **1530** in these examples.

(80) At operation **1710**, the copilot may construct a second prompt to the machine-learned large language model, the second prompt including a second portion of the live transcript of the network-based communication session, and the first summary (e.g., the response received at operation **1525**) and second instructions to summarize both the second portion of the live transcript and the summarization. In some examples, the first and second portions of the transcript may be overlapping. Using overlapping transcripts may reduce incidents where the final summary produced by the summary of summaries approach fails to provide appropriate relevant context.

(81) At operation **1715**, the copilot may submit the second prompt to the machine-learned large language model. At operation **1720**, the copilot may receive a second response from the machine-learned large language model, the second response being for the second prompt. As noted for FIG. **15**, the prompts and responses may be sent and received locally or via a network.

(82) At operation **1740**, the copilot determines the result based upon the second response. In some examples, the second response may be the result. In other examples, the system may utilize the second result in another prompt to the LLM—for example, issuing a query on the summary and the response to that prompt may be the result. In these examples, the prompts to create the initial summaries may include the query and ask the LLM to summarize the transcript and include relevant information about the query. While FIG. **17** illustrates the use of two prompts and two live transcript portions, the transcript may be broken up into more than two portions and additional prompts may be utilized.

(83) FIG. **18** illustrates a flowchart of a method **1800** of querying a transcript of a network-based communication session according to some examples of the present disclosure. The method **1800** may operate upon a transcript created using a summary of summaries or rolling summary method where the prompt includes the query and instructions to preserve details about the query. The

method **1800** may operate as part of method **1500**, such as after one or more of FIG. **16** or **17**. At operation **1810**, the copilot may construct a query-answer prompt that asks the LLM to answer the query using the summary. At operation **1815**, the copilot may submit the query-answer prompt. As previously noted, this may be locally or remotely sent. At operation **1820**, the copilot may receive the result. In some examples, the result may be used with operation **1530** of FIG. **15**.

(84) FIG. **19** illustrates a flowchart of a method **1900** for caching a predicted query response according to some examples of the present disclosure. At operation **1910**, the system may construct a prompt, e.g., based upon a query, a summarization task, or the like. At operation **1915**, the copilot may submit the prompt. The prompt may be created at operation **1910** and submitted at operation **1915** based upon a submitted user query, the prompt may also (in addition to requesting the answer to the query) include a request for the large language model to predict a follow up query and provide that with the response. At operation **1920**, the response may be received including the predicted follow up. At operation **1925**, the response may be provided to the user, e.g., by providing it in a GUI, such as a GUI of the network-based communication service. At operation **1930**, the copilot may, without a prompt from the user asking the follow up query, construct a second prompt for the LLM to answer the follow up query and submit the prompt to the LLM at operation **1935**. The response may be received and stored (e.g., in a volatile or non-volatile memory) at operation **1940**. In some examples, the prompt constructed at operation **1910** may also request that the LLM answer the query in the first prompt and provide the response with the response received at operation **1920**. For these examples, operations **1930**, **1935**, and **1940** may be skipped.

(85) At operation **1945** a later query may be received from the participant. At operation **1950** it is determined if the query corresponds to a predicted follow up. In some examples, multiple follow up questions may be requested from the LLM and stored before the participant asks. If the query does not correspond to a predicted follow up query stored by the system, then at operation **1960**, the system may handle the query normally. For example, by one or more of the method in FIGS. **15-18**.

(86) In some examples, similarity may be measured based upon an exact match between the query and the predicted follow up query. For example, the system may display the LLM's predicted follow up question to the user and ask the user if they would like the response. The user may indicate that they wish to see the answer. In these examples, the query issued at operation **1945** is identical to the predicted follow up.

(87) In other examples, various semantic similarity algorithms may be used to determine a similarity between the later query and the predicted follow up. For example, by using a latent semantic analysis and the like. If the submitted query and the predicted follow up are within a specified similarity, then they may be deemed to be corresponding at operation **1950**. If the later query corresponds to the predicted follow up at operation **1950**, then at operation **1955**, the stored response may be provided to the user, e.g., in the GUI.

(88) As previously noted, the copilot may utilize one or more machine-learning algorithms, for example, to answer questions about the communication session, produce prompts for other models, determine suggested questions, and the like. FIG. **20** shows an example machine learning module **2000** according to some examples of the present disclosure. The machine learning module **2000** may be implemented in whole or in part by one or more computing devices. In some examples, the training module **2010** may be implemented by a different device than the prediction module **2020**. In these examples, the model **2080** may be created on a first machine and then sent to a second machine. In some examples, one or more portions of the machine learning module **2000** may be implemented by network-based communication service **1214** or network-based language model service **1216** of FIG. **12**.

(89) In some examples, machine learning module **2000** utilizes a training module **2010** and a prediction module **2020**. Training module **2010** inputs training feature data **2030** into selector module **2050**. The training feature data **2030** may include one or more sets of training data. The

training feature data **2030** may be labeled with the desired output. In other examples, the training data may not be labeled, and the model may be trained using unsupervised methods and/or feedback data-such as through a reinforcement learning method. The feedback data may be a measure of error between a desired result of the algorithm and the actual result.

(90) Selector module **2050** converts and/or selects training vector **2060** from the training feature data **2030**. For example, the selector module **2050** may filter, select, transform, process, or otherwise convert the training data. For example, the selector module **2050** may apply one or more feature selection algorithms to find features in the training data. The selected data may fill training vector **2060** and comprises a set of the training data that is determined to be predictive of a result. Information chosen for inclusion in the training vector **2060** may be all the training feature data **2030** or in some examples, may be a subset of all the training feature data **2030**. Selector module **2050** may also convert or otherwise process the training feature data **2030** such as normalization, encoding, and the like. The training vector **2060** may be utilized (along with any applicable labels) by the machine learning algorithm **2070** to produce a model **2080**. In some examples, other data structures other than vectors may be used. The machine learning algorithm **2070** may learn one or more layers of a model. Example layers may include convolutional layers, dropout layers, pooling/up sampling layers, SoftMax layers, and the like. Example models may be a neural network, where each layer is comprised of a plurality of neurons that take a plurality of inputs, weight the inputs, input the weighted inputs into an activation function to produce an output which may then be sent to another layer. Example activation functions may include a Rectified Linear Unit (ReLU), and the like. Layers of the model may be fully or partially connected.

(91) In the prediction module **2020**, feature data **2090** is input to the selector module **2095**. Selector module **2095** may operate the same, or differently than selector module **2050**. In some examples, selector modules **2050** and **2095** are the same modules or different instances of the same module. Selector module **2095** produces vector **2097**, which is input into the model **2080** to produce an output **2099**. For example, the weightings and/or network structure learned by the training module **2010** may be executed on the vector **2097** by applying vector **2097** to a first layer of the model **2080** to produce inputs to a second layer of the model **2080**, and so on until the encoding is output. As previously noted, other data structures may be used other than a vector (e.g., a matrix).

(92) The training module **2010** may operate in an offline manner to train the model **2080**. The prediction module **2020**, however, may be designed to operate in an online manner. It should be noted that the model **2080** may be periodically updated via additional training and/or user feedback. For example, additional training feature data **2030** may be collected as users provide feedback on the performance of the predictions.

(93) The machine learning algorithm **2070** may be selected from among many different potential supervised or unsupervised machine learning algorithms. Examples of learning algorithms include artificial neural networks, Generative Pretrained Transformer (GPT), convolutional neural networks, Bayesian networks, instance-based learning, support vector machines, decision trees (e.g., Iterative Dichotomiser 3, C4.5, Classification and Regression Tree (CART), Chi-squared Automatic Interaction Detector (CHAID), and the like), random forests, linear classifiers, quadratic classifiers, k-nearest neighbor, k-means, linear regression, logistic regression, a region based CNN, a full CNN (for semantic segmentation), a mask R-CNN algorithm for instance segmentation, Latent Dirichlet Algorithm (LDA), and hidden Markov models. Examples of unsupervised learning algorithms include expectation-maximization algorithms, vector quantization, and information bottleneck method.

(94) While one training phase and a single model are shown, in other examples, multiple training phases may be used. For example, a supervised training phase may be used to initially train an initial model and an unsupervised phase may be used to finalize the model.

(95) As noted, the machine-learning model may be used to answer questions about the communication session. The model **2080** in these examples may be a large language model (LLM)

such as a GPT model. In these examples, a first learning phase may utilize training feature data **2030** that is demonstration data curated by human labelers. The model **2080** is a supervised policy (e.g., a Supervised Fine-Tuning (SFT) model) that generates outputs from feature data **2090** that comprises a selected list of prompts. Next, a large number of SFT model outputs are voted on by human labelers to create a new dataset comprising comparison data. This comparison data is then used as training feature data **2030** to create a new model **2080** called a reward model. The reward model is then used to further fine-tune the SFT model to output a policy model **2080**. In some examples, the training process may utilize feature data specific to network-based communication sessions.

(96) As additionally noted, a machine-learning model may be used to produce prompts for the LLM. In these examples, the training feature data **2030** may be queries labeled with the proper prompt and the output model **2080** may be a human language prompt for the LLM. The feature data **2090** may then be live queries issued by users and the output may be the proper prompt.

(97) Finally, the machine-learning model may be used to determine suggested questions based upon the network-based communication session content and context. In these examples, the training feature data **2030** may be communication session transcripts, participant information, and other context labeled with suggested questions. The feature data **2090** may be actual content and context data and the output **2099** may be an identifier of a suggested question. In some examples, prior to displaying suggested questions, the system may generate one or more prompts for the suggested questions and issue them to the LLM to obtain and cache a response as shown in FIG. **19**.

(98) FIG. **21** illustrates a block diagram of an example machine **2100** upon which any one or more of the techniques (e.g., methodologies) discussed herein may be performed. In alternative embodiments, the machine **2100** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine **2100** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **2100** may act as a peer machine in peer-to-peer (P2P) (or other distributed) network environment. The machine **2100** may be in the form of a server computing system (e.g., that operates a service), a laptop, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a smart phone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), other computer cluster configurations.

(99) In some examples, machine **2100** may render, or create one or more of the GUIs of FIGS. **1-11** & **14**; implement any of the computing devices of FIG. **12**; implement the components of FIGS. **13** & **20**; and implement the methods of FIGS. **15-19**. In some examples, the machine **2100** may accomplish the rendering, creation, or implementation of the previous FIGS. by specific hardware configurations, software configuring general purpose hardware, or a combination of specific hardware and software.

(100) Examples, as described herein, may include, or may operate on one or more logic units, components, or mechanisms (“components”). Components are tangible entities (e.g., hardware) capable of performing specified operations and may be configured or arranged in a certain manner. In an example, circuits may be arranged (e.g., internally or with respect to external entities such as other circuits) in a specified manner as a component. In an example, the whole or part of one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware processors may be configured by firmware or software (e.g., instructions, an application portion, or an application) as a component that operates to perform specified operations. In an example, the software may reside on a machine readable medium. In an example, the software,

when executed by the underlying hardware of the component, causes the hardware to perform the specified operations of the component.

(101) Accordingly, the term “component” is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which component are temporarily configured, each of the components need not be instantiated at any one moment in time. For example, where the components comprise a general-purpose hardware processor configured using software, the general-purpose hardware processor may be configured as respective different components at different times. Software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different component at a different instance of time.

(102) Machine (e.g., computer system) **2100** may include one or more hardware processors, such as processor **2102**. Processor **2102** may be a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof. Machine **2100** may include a main memory **2104** and a static memory **2106**, some or all of which may communicate with each other via an interlink (e.g., bus) **2108**. Examples of main memory **2104** may include Synchronous Dynamic Random-Access Memory (SDRAM), such as Double Data Rate memory, such as DDR4 or DDR5. Interlink **2108** may be one or more different types of interlinks such that one or more components may be connected using a first type of interlink and one or more components may be connected using a second type of interlink. Example interlinks may include a memory bus, a peripheral component interconnect (PCI), a peripheral component interconnect express (PCIe) bus, a universal serial bus (USB), or the like.

(103) The machine **2100** may further include a display unit **2110**, an alphanumeric input device **2112** (e.g., a keyboard), and a user interface (UI) navigation device **2114** (e.g., a mouse). In an example, the display unit **2110**, input device **2112** and UI navigation device **2114** may be a touch screen display. The machine **2100** may additionally include a storage device (e.g., drive unit) **2116**, a signal generation device **2118** (e.g., a speaker), a network interface device **2120**, and one or more sensors **2121**, such as a global positioning system (GPS) sensor, compass, accelerometer, or other sensor. The machine **2100** may include an output controller **2128**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.).

(104) The storage device **2116** may include a machine readable medium **2122** on which is stored one or more sets of data structures or instructions **2124** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **2124** may also reside, completely or at least partially, within the main memory **2104**, within static memory **2106**, or within the hardware processor **2102** during execution thereof by the machine **2100**. In an example, one or any combination of the hardware processor **2102**, the main memory **2104**, the static memory **2106**, or the storage device **2116** may constitute machine readable media.

(105) While the machine readable medium **2122** is illustrated as a single medium, the term “machine readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **2124**.

(106) The term “machine readable medium” may include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **2100** and that cause the machine **2100** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Non-limiting machine readable medium examples may include solid-state memories, and optical and magnetic media. Specific examples of machine readable media may include: non-volatile memory,

such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; Random Access Memory (RAM); Solid State Drives (SSD); and CD-ROM and DVD-ROM disks. In some examples, machine readable media may include non-transitory machine readable media. In some examples, machine readable media may include machine readable media that is not a transitory propagating signal.

(107) The instructions **2124** may further be transmitted or received over a communications network **2126** using a transmission medium via the network interface device **2120**. The Machine **2100** may communicate with one or more other machines wired or wirelessly utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol (IP), transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communication networks may include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), Plain Old Telephone (POTS) networks, and wireless data networks such as an Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, an IEEE 802.15.4 family of standards, a 5G New Radio (NR) family of standards, a Long Term Evolution (LTE) family of standards, a Universal Mobile Telecommunications System (UMTS) family of standards, peer-to-peer (P2P) networks, among others. In an example, the network interface device **2120** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **2126**. In an example, the network interface device **2120** may include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. In some examples, the network interface device **2120** may wirelessly communicate using Multiple User MIMO techniques.

EXAMPLES

(108) Example 1 is a system providing a personalized assistant within a network-based communication session, the system comprising: a processor; and a memory storage device storing instructions thereon, which, when executed by the processor, cause the system to perform operations comprising: determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the network-based communication session after the threshold duration of time: obtaining content associated with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session; creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output, a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; transforming the request data to a command based upon a command template; providing the command to a pre-trained generative language model with the request data; receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

(109) In Example 2, the subject matter of Example 1 includes, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another

communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

(110) In Example 3, the subject matter of Examples 1-2 includes, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

(111) In Example 4, the subject matter of Examples 1-3 includes, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session; and causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(112) In Example 5, the subject matter of Examples 1-4 includes, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for each segment of the plurality of segments, providing as input to the pre-trained generative language model content from the segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session, for each segment of the plurality of segments; providing to the pre-trained generative language model a final input, the final input including i) the summary description of the network-based communication session, as output by the generative language model for each segment of the plurality of segments and ii) an instruction to generate an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; responsive to providing the final input to the generative language model, receiving as output from the generative language model an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; and causing the overall summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(113) In Example 6, the subject matter of Examples 1-5 includes, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for a first segment in the plurality of segments, providing as input to the pre-trained generative language

model i) content from the first segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content from the first segment; for each segment in the plurality of segments subsequent to the first segment, providing as input to the pre-trained generative language model i) the summary description of the network-based communication session output by the pre-trained generative language model based on a prior segment and content from the segment, and ii) an instruction to generate a summary description of the network-based communication session; receiving as output from the generative language model a final summary description of the network-based communication session, based on the generative language model processing a final prompt for a last segment in the plurality of segments; and causing the final summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(114) In Example 7, the subject matter of Examples 1-6 includes, wherein the content associated with the in-session network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: determine existence of a specific type of relationship between the first communication session participant and a second communication session participant; extracting from the content one or more content items representing a communication made by the second communication session participant; providing as input to a pre-trained generative language model, i) the one or more extracted content items, and ii) an instruction to generate a summary description of communications made by the second communication session participant; receiving as output from the generative language model the summary description of communications made by the second communication session participant; and causing the summary description of communications made by the second communication session participant to be presented to the first communication session participant in a graphical user interface of the network-based communication session.

(115) In Example 8, the subject matter of Examples 1-7 includes, wherein the pre-trained generative language model has been fine-tuned, using a supervised learning technique, to generate suggested utterances not present in a portion of content based on a conversation of communication session participants as expressed in content, wherein a training dataset used in fine-tuning the pre-trained generative language model includes a plurality of instances of a communication session transcript having a plurality of chronologically ordered content items, each content item in the plurality of chronologically ordered content items representing a communication made by a communication session participant, wherein one or more communications have been labeled as questions, statements or opinions.

(116) Example 9 is a method for providing a personalized assistant within a network-based communication session, the method comprising: determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the network-based communication session after the threshold duration of time: obtaining content associated with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session; creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output,

a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; transforming the request data to a command based upon a command template; providing the command to a pre-trained generative language model with the request data; receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

(117) In Example 10, the subject matter of Example 9 includes, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

(118) In Example 11, the subject matter of Examples 9-10 includes, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

(119) In Example 12, the subject matter of Examples 9-11 includes, providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session; and causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(120) In Example 13, the subject matter of Examples 9-12 includes, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for each segment of the plurality of segments, providing as input to the pre-trained generative language model content from the segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session, for each segment of the plurality of segments; providing to the pre-trained generative language model a final input, the final input including i) the summary description of the network-based communication session, as output by the generative language model for each segment of the plurality of segments and ii) an instruction to generate an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; responsive to providing the final input to the generative language model, receiving as output from the generative language model an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; and causing the overall summary description of the network-based

communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(121) In Example 14, the subject matter of Examples 9-13 includes, segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for a first segment in the plurality of segments, providing as input to the pre-trained generative language model i) content from the first segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content from the first segment; for each segment in the plurality of segments subsequent to the first segment, providing as input to the pre-trained generative language model i) the summary description of the network-based communication session output by the pre-trained generative language model based on a prior segment and content from the segment, and ii) an instruction to generate a summary description of the network-based communication session; receiving as output from the generative language model a final summary description of the network-based communication session, based on the generative language model processing a final prompt for a last segment in the plurality of segments; and causing the final summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(122) In Example 15, the subject matter of Examples 9-14 includes, wherein the content associated with the in-session network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the method further comprises: determining existence of a specific type of relationship between the first communication session participant and a second communication session participant; extracting from the content one or more content items representing a communication made by the second communication session participant; providing as input to a pre-trained generative language model, i) the one or more extracted content items, and ii) an instruction to generate a summary description of communications made by the second communication session participant; receiving as output from the generative language model the summary description of communications made by the second communication session participant; and causing the summary description of communications made by the second communication session participant to be presented to the first communication session participant in a graphical user interface of the network-based communication session.

(123) In Example 16, the subject matter of Examples 9-15 includes, wherein the pre-trained generative language model has been fine-tuned, using a supervised learning technique, to generate suggested utterances not present in a portion of content based on a conversation of communication session participants as expressed in content, wherein a training dataset used in fine-tuning the pre-trained generative language model includes a plurality of instances of a communication session transcript having a plurality of chronologically ordered content items, each content item in the plurality of chronologically ordered content items representing a communication made by a communication session participant, wherein one or more communications have been labeled as questions, statements or opinions.

(124) Example 17 is a system providing a personalized assistant within a network-based communication session, the system comprising: means for determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the network-based communication session after the threshold duration of time: means for obtaining content associated

with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session, means for creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output, a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; means for transforming the request data to a command based upon a command template; means for providing the command to a pre-trained generative language model with the request data; means for receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and means for causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

(125) In Example 18, the subject matter of Example 17 includes, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

(126) In Example 19, the subject matter of Examples 17-18 includes, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

(127) In Example 20, the subject matter of Examples 17-19 includes, means for providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; means for receiving as output from the generative language model a summary description of the network-based communication session; and means for causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

(128) Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.

(129) Example 22 is an apparatus comprising means to implement any of Examples 1-20.

(130) Example 23 is a system to implement any of Examples 1-20.

(131) Example 24 is a method to implement any of Examples 1-20.

Claims

1. A system providing a personalized assistant within a network-based communication session, the system comprising: a processor; and a memory storage device storing instructions thereon, which, when executed by the processor, cause the system to perform operations comprising: determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the

network-based communication session after the threshold duration of time: obtaining content associated with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session; creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output, a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; transforming the request data to a command based upon a command template; providing the command to a pre-trained generative language model with the request data; receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

2. The system of claim 1, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

3. The system of claim 1, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

4. The system of claim 1, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session; and causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

5. The system of claim 1, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for each segment of the plurality of segments, providing as input to the pre-trained generative language model content from the segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session, for each segment of the plurality of segments; providing to the pre-trained generative language model a final input, the final input including i) the summary description of the network-based communication session, as output by the generative language model for each segment of the plurality of segments and ii) an instruction to generate an overall summary description of the network-based communication

session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; responsive to providing the final input to the generative language model, receiving as output from the generative language model an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; and causing the overall summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

6. The system of claim 1, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for a first segment in the plurality of segments, providing as input to the pre-trained generative language model i) content from the first segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content from the first segment; for each segment in the plurality of segments subsequent to the first segment, providing as input to the pre-trained generative language model i) the summary description of the network-based communication session output by the pre-trained generative language model based on a prior segment and content from the segment, and ii) an instruction to generate a summary description of the network-based communication session; receiving as output from the generative language model a final summary description of the network-based communication session, based on the generative language model processing a final prompt for a last segment in the plurality of segments; and causing the final summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

7. The system of claim 1, wherein the content associated with the in-session network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: determine existence of a specific type of relationship between the first communication session participant and a second communication session participant; extracting from the content one or more content items representing a communication made by the second communication session participant; providing as input to a pre-trained generative language model, i) the one or more extracted content items, and ii) an instruction to generate a summary description of communications made by the second communication session participant; receiving as output from the generative language model the summary description of communications made by the second communication session participant; and causing the summary description of communications made by the second communication session participant to be presented to the first communication session participant in a graphical user interface of the network-based communication session.

8. The system of claim 1, wherein the pre-trained generative language model has been fine-tuned, using a supervised learning technique, to generate suggested utterances not present in a portion of content based on a conversation of communication session participants as expressed in content, wherein a training dataset used in fine-tuning the pre-trained generative language model includes a plurality of instances of a communication session transcript having a plurality of chronologically

ordered content items, each content item in the plurality of chronologically ordered content items representing a communication made by a communication session participant, wherein one or more communications have been labeled as questions, statements or opinions.

9. A method for providing a personalized assistant within a network-based communication session, the method comprising: determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the network-based communication session after the threshold duration of time: obtaining content associated with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session; creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output, a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; transforming the request data to a command based upon a command template; providing the command to a pre-trained generative language model with the request data; receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

10. The method of claim 9, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

11. The method of claim 9, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

12. The method of claim 9, further comprising: providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session; and causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

13. The method of claim 9, wherein the memory storage device is storing instructions, which, when executed by the one or more processors, cause the system to perform additional operations comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model; using the generative language model to generate a summary description of the network-based communication session, by: for each segment of the plurality of segments, providing as input to the pre-trained generative language model content from the segment, and ii) an instruction to generate a summary description of the network-based

communication session, based on the content; receiving as output from the generative language model a summary description of the network-based communication session, for each segment of the plurality of segments; providing to the pre-trained generative language model a final input, the final input including i) the summary description of the network-based communication session, as output by the generative language model for each segment of the plurality of segments and ii) an instruction to generate an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; responsive to providing the final input to the generative language model, receiving as output from the generative language model an overall summary description of the network-based communication session, based on the summary description of the network-based communication session as output by the generative language model for each segment of the plurality of segments; and causing the overall summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

14. The system of claim 9, further comprising: segmenting the content into a plurality of segments, each segment in the plurality of segments having a size that is based on a maximum input size requirement of the pre-trained generative language model: using the generative language model to generate a summary description of the network-based communication session, by: for a first segment in the plurality of segments, providing as input to the pre-trained generative language model i) content from the first segment, and ii) an instruction to generate a summary description of the network-based communication session, based on the content from the first segment; for each segment in the plurality of segments subsequent to the first segment, providing as input to the pre-trained generative language model i) the summary description of the network-based communication session output by the pre-trained generative language model based on a prior segment and content from the segment, and ii) an instruction to generate a summary description of the network-based communication session; receiving as output from the generative language model a final summary description of the network-based communication session, based on the generative language model processing a final prompt for a last segment in the plurality of segments; and causing the final summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.

15. The method of claim 9, wherein the content associated with the in-session network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the method further comprises: determining existence of a specific type of relationship between the first communication session participant and a second communication session participant; extracting from the content one or more content items representing a communication made by the second communication session participant; providing as input to a pre-trained generative language model, i) the one or more extracted content items, and ii) an instruction to generate a summary description of communications made by the second communication session participant; receiving as output from the generative language model the summary description of communications made by the second communication session participant; and causing the summary description of communications made by the second communication session participant to be presented to the first communication session participant in a graphical user interface of the network-based communication session.

16. The method of claim 9, wherein the pre-trained generative language model has been fine-tuned, using a supervised learning technique, to generate suggested utterances not present in a portion of

content based on a conversation of communication session participants as expressed in content, wherein a training dataset used in fine-tuning the pre-trained generative language model includes a plurality of instances of a communication session transcript having a plurality of chronologically ordered content items, each content item in the plurality of chronologically ordered content items representing a communication made by a communication session participant, wherein one or more communications have been labeled as questions, statements or opinions.

17. A system providing a personalized assistant within a network-based communication session, the system comprising: means for determining a first communication session participant has joined a network-based communication session after a threshold duration of time subsequent to the start time of the network-based communication session; and responsive to determining the first communication session participant has joined the network-based communication session after the threshold duration of time: means for obtaining content associated with the network-based communication session, the content originating during a window of time spanning the start time of the network-based communication session and the time at which the communication session participant joined the network-based communication session; means for creating request data for a pre-trained generative language model based upon a portion of the content associated with the network-based communication session and an instruction requesting, as output, a predetermined number of suggested utterances not present in the portion of the content, each utterance relating to one or more topics corresponding to the portion of the content; means for transforming the request data to a command based upon a command template; means for providing the command to a pre-trained generative language model with the request data; means for receiving a response from the generative language model, the response including the predetermined number of suggested utterances; and means for causing one or more of the predetermined number of suggested utterances to be presented to the communication session participant in a graphical user interface of the network-based communication session, while the network-based communication session is in session.

18. The system of claim 17, wherein the instruction requesting, as output, the predetermined number of suggested utterances not present in the portion of the content includes a request to generate questions that have not already been asked by another communication session participant, or a request to generate questions that exclude questions that have already been asked by another communication session participant.

19. The system of claim 17, wherein the content associated with the network-based communication session is structured as a communication session transcript having a plurality of chronologically ordered content items, wherein each content item in the plurality of chronologically ordered content items represents a communication made by a communication session participant and includes data indicating the name of the communication session participant who made the communication, wherein the instruction to generate the predetermined number of suggested utterances not present in the portion of the content includes a request to include the name of a communication session participant to whom a question should be directed.

20. The system of claim 17, further comprising: means for providing as a second input to the pre-trained generative language model an instruction to generate a summary description of the network-based communication session, based on the content; means for receiving as output from the generative language model a summary description of the network-based communication session; and means for causing the summary description of the network-based communication session to be presented to the communication session participant in a graphical user interface of the network-based communication session.
